Pavan Mantrigragada

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Education

University of Maryland College Park — College Park, MD
PhD Computer Science — *Expected December 2027*

University of Maryland College Park — College Park, MD
M.Eng. Robotics — *December 2022*

Indian Institute of Technology Hyderabad — Hyderabad, India
B.Tech Mechanical Engineering (Honors) — *May 2017*

Technical Skills

Languages: Python, C++, C
Machine Learning: PyTorch, Accelerate, Transformers, Diffusers, Weights & Biases, multi-GPU training, SLURM
3D Vision: PyTorch3D, Open3D, OpenCV, Nerf Studio, LichtFeld Studio
Robotics: ROS, MoveIt, Eigen, Gym
Simulation & Design: Isaac Sim, Pybullet, MuJoCo, Blender, Onshape
Hardware: Shadow Hand, Franka, UR, KUKA LBR iiwa, Robotiq grippers, Clearpath Moose, Raspberry Pi
Tools: Docker, Git, MATLAB, LaTeX

Research

- Dessalene, E., He, B., Maynord, M., Tussa, Y., Mantripragada, P., Karabatis, Y., Roy, N. and Aloimonos, Y., 2026, September. Feel (force-enhanced egocentric learning): A dataset for physical action understanding. In European Conference on Computer Vision (pp. 546-563). Cham: Springer Nature Switzerland.

- Burner, L., Mantripragada, P., Caddeo, G.M., Natale, L., Fermüller, C. and Aloimonos, Y., 2024. Extremum Seeking Controlled Wiggling for Tactile Insertion. arXiv preprint arXiv:2410.02595. IROS 2026.

- Mantripragada, P., Deshmukh, S., Dessalene, E., Desai, M. and Aloimonos, Y., 2026. Monocular Reconstruction of Neural Tactile Fields. arXiv preprint arXiv:2602.12508.

- Dessalene, E., Mantripragada, P., Maynord, M. and Aloimonos, Y., 2025. EmbodiSwap for Zero-Shot Robot Imitation Learning. arXiv preprint arXiv:2510.03706.

- Paralikar, A., Mantripragada, P., Nguyen, T., Arjoune, Y., Shekhar, R. and Monfaredi, R., 2025. Robot-assisted ultrasound probe calibration for image-guided interventions. International journal of computer assisted radiology and surgery, 20(5), pp.859-868.

- Shahidzadeh, A.H., Yoo, S.J., Mantripragada, P., Singh, C.D., Fermüller, C. and Aloimonos, Y., 2024, May. Actexplore: Active tactile exploration on unknown objects. In 2024 IEEE International Conference on Robotics and Automation (ICRA) (pp. 3411-3418). IEEE.

- Fischer, E., Mantripragada, P., Espinoza, A.K., Yarmolenko, P., Cleary, K., Sharma, K. and Monfaredi, R., 2023, April. Feasibility of a novel CT grid template for needle-based targeting in a phantom. In Medical Imaging 2023: Image-Guided Procedures, Robotic Interventions, and Modeling (Vol. 12466, pp. 404-408). SPIE.

- Ganguly, K., Mantripragada, P., Parameshwara, C.M., Fermüller, C., Sanket, N.J. and Aloimonos, Y., 2022. Gradtac: Spatio-temporal gradient based tactile sensing. Frontiers in Robotics and AI, 9, p.898075.

- Ganguly, K., Sadrfaridpour, B., Mantripragada, P., Sanket, N.J., Fermüller, C. and Aloimonos, Y., 2020. Grasping in the dark: Zero-shot object grasping using tactile feedback. arXiv preprint arXiv:2011.00712.

Other Projects

Closed-Loop Control of Spraying Actions Using Vision Aug 2023 – Apr 2024

- Collaborated with the Army Research Laboratory to develop a Newton-method based IK solver tailored for the UR16 robotic arm. This solver is used for autonomous decontamination of vehicles in the wild.
- Unlike current methods, our proposed formulation can effectively manage singularities, kinematic redundancies, and discontinuities in joint trajectories, all while maintaining a latency of 3ms per query.

Improving VQA Accuracy Using Answer Equivalence Aug 2023 – Dec 2023

- Performed a comprehensive analysis to assess the impact of Answer Equivalence on visual question answering (VQA) models.
- Discovered that integrating BERT matching in evaluation enhances the VQA accuracy of state-of-the-art models like Prophet, BEiT3, and PALI-X-VPD by a minimum of 5%.

Dexterous Manipulation under Constraints Sep 2021 – Dec 2021

- Created a ROS packaged for 30 DOF manipulator-gripper (UR10 & ShadowHand) assembly which is used grasping objects.
- Designed a joint control-policy based on contact feedback to achieve stable grasps by an anthropomorphic robot hand.
- Demonstrated this grasping technique on grasping door handles and successfully opening the doors.
- A custom Inverse Kinematic solver was used to generate the trajectory of the hand for opening the door.

Learning high-level policies for model predictive control Sep 2021 – Nov 2021

- Implemented an MPC to control robots which takes high-level policy parameters from a neural network based on robot observations.
- This algorithm is demonstrated for a quadrotor traversing through a fast-swinging gate.

Potential Functions based Sampling Heuristic for Optimal Path Planning Jan 2021 – May 2021

- Demonstrated the use of potential functions in selective sampling of states in RRT* algorithm. This algorithm is designed to escape local minima problem traditional APF based planners face.
- We found that this algorithm can generate paths with far less samples than RRT* requires.

Work Experience

Research Intern Jun 2023 – Aug 2023

School of Medicine, UM *Baltimore, United States*

- Developed an android AR application for phantom-limb pain mirror therapy.
- Integrated visual human pose estimation from a mobile camera with an application on Meta Quest2 using OSC protocol

Research Intern Mar 2022 – Aug 2022

Children's National Health System *DC, United States*

- Designed an experimental setup for a robotic-assisted ultrasound probe calibration for AR-guided surgeries
- Evaluated accuracy and precision of 2D & 3D reconstruction loss of our calibration phantom on a cross-wire phantom.
- Co-authored a study on novel grid-marker fiducial to assist needle biopsy training.

Junior Research Fellow Aug 2019 – May 2020

Indian Institute of Technology Bombay *Bombay, India*

- Designed a powered lower-limb exoskeleton for rehabilitation of patients with walking disabilities, using an assist-as-needed force field controller.
- Proposed a novel controller using a combination of velocity and force field controllers to decouple the gait-speed and maximum applied actuator forces.

Maintenance Officer Jun 2017 – Jul 2019

Bharat Petroleum *Bombay, India*

- Led equipment reliability and upkeep operations for the Fluid Catalytic Cracker Unit at Mumbai Refinery.
- Performed machinery diagnostics and condition-monitoring analysis to predict equipment failures.